Artificial Intelligence 2: Bayes' Theorem

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Outline of Topics

Tossing biased coins

Example: Suppose we have some biased coins that have two possible biases of $\theta_{0.9} = 0.9$ and $\theta_{0.1} = 0.1$. The probability of a coin have the bias $\theta_{0.9}$ is

$$p(\theta_{0.9}) = c.$$

We observe the probability of that a coin will land heads up is

$$p(x_h) = a,$$

and the likelihood

$$p(x_h|\theta_{0.9}) = d$$

Question: Derive the posterior probability $p(\theta_{0.9}|x_h)$ without using Bayes' theorem.

Tossing a biased coin

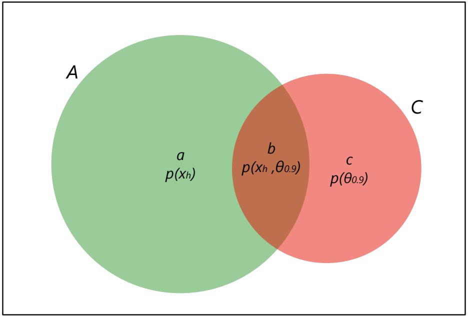


Figure 1: Bayes' theorem from a Venn Diagram. The total area of the whole box is 1. The area a of the green disk A represents the probability $p(x_h)$ that a coin will land with heads, x_h. The area c of the pink disk C represents the probability $p(\theta_{0.9})$ that a coin has bias $\theta = 0.9$. The overlap between A and C is the area b, which represents the joint probability $p(x_h, \theta_{0.9})$ that a coin will land heads up and that a coin has bias $\theta_{0.9}$

Bayes' theorem from Venn Diagram

From disk A, we can obtain

$$p(\theta_{0.9}|x_h) = \frac{b}{a} = \frac{p(x_h, \theta_{0.9})}{p(x_h)}, \tag{1}$$

and from disk C we have:

$$p(x_h|\theta_{0.9}) = \frac{b}{c} = \frac{p(x_h, \theta_{0.9})}{p(\theta_{0.9})} \tag{2}$$

From the above two equations, we have

$$p(x_h, \theta_{0.9}) = p(\theta_{0.9}|x_h)p(x_h) \tag{3}$$

$$p(x_h, \theta_{0.9}) = p(x_h|\theta_{0.9})p(\theta_{0.9}), \tag{4}$$

which yield

$$p(\theta_{0.9}|x_h)p(x_h) = p(x_h|\theta_{0.9})p(\theta_{0.9}) \Rightarrow p(\theta_{0.9}|x_h) = \frac{p(x_h|\theta_{0.9})p(\theta_{0.9})}{p(x_h)} \tag{5}$$

Example: A doctor's patient record

	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8	θ_9	θ_{10}	Sum
x_1	8	9	9	5	4	1	1	0	0	0	37
x_2	3	5	8	9	14	10	3	3	0	0	55
x_3	0	1	1	10	16	11	12	7	8	5	71
x_4	0	0	1	0	3	5	10	7	7	4	37
Sum	11	15	19	24	37	27	26	17	15	9	200

Table 1: Joint observations of symptoms x_r, i.e., rows $r = \{1, 2, 3, 4\}$ and diseases θ_c, i.e., columns $c = \{1, 2, \ldots, 10\}$. The number $n(x_r, \theta_C)$ in each cell represents the number of patients with the symptoms x_r and disease θ_c.

Doctor's Questions

Questions from the doctor:

- Q1: What is the joint probability $p(x_3, \theta_2)$ that a patient has the symptom x_3 and the disease θ_2?
- Q2: What is the probability $p(x_3)$ that a patient has the symptom x_3?
- Q3: What is the probability $p(\theta_2)$ that a patient has the disease θ_2?
- Q4: What is the conditional probability $p(x_3|\theta_2)$ that a patient has the symptom x_3 given that he has the disease θ_2?
- Q5: What is the conditional probability $p(\theta_2|x_3)$ that a patient has the disease θ_2 give that he has the symptom x_3?

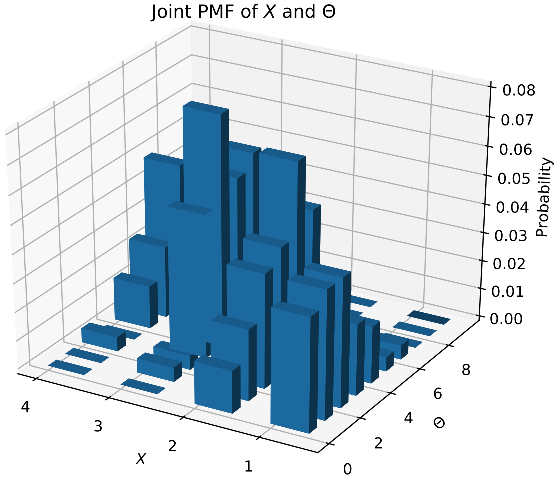
Random Variables for doctors

We first use random variables to model the patient record: let the discrete random variable X to represent the row number, i.e., one of the four symptoms, of which the range $R_X = \{x_1, x_2, x_3, x_4\}$ and Θ to represent the column number, i.e., one of the 10 diseases, of which the range $R_\Theta = \{\theta_1, \theta_2, \ldots, \theta_{10}\}$. We can now calculate the joint probability and obtain the joint PMF of X and Θ as the following table

	θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8	θ_9	θ_{10}	$P(X)$
x_1	0.04	0.045	0.045	0.025	0.02	0.005	0.005	0	0	0	0.185
x_2	0.015	0.025	0.04	0.045	0.07	0.05	0.015	0.015	0	0	0.275
x_3	0	0.005	0.005	0.05	0.08	0.055	0.06	0.035	0.04	0.025	0.355
x_4	0	0	0.005	0	0.015	0.025	0.05	0.035	0.035	0.02	0.185
$P(\Theta)$	0.055	0.075	0.095	0.12	0.185	0.135	0.13	0.085	0.075	0.045	1

Table 2: Joint PMF of two discrete random variables X and Θ and their marginal PMFs. The marginal PMFs $P(X)$ and $P(\Theta)$ are also know as **the marginal likelihood** (of symptoms) and **the prior distribution** (of disease), respectively.

Joint PMF of X and Θ



Doctor's Questions

- Q1: What is the joint probability $p(x_3, \theta_2)$ that a patient has the symptom x_3 and the disease θ_2? A1: $p(x_3, \theta_2) = 0.005$
- Q2: What is the probability $p(x_3)$ that a patient has the symptom x_3? $p(x_3) = 0.355$
- Q3: What is the probability $p(\theta_2)$ that a patient has the disease θ_2? $p(\theta_2) = 0.075$
- Q4: What is the conditional probability $p(x_3|\theta_2)$ that a patient has the symptom x_3 given that he has the disease θ_2?
  A4:
  $$p(x_3|\theta_2) = \frac{p(x_3, \theta_2)}{p(\theta_2)} = \frac{0.005}{0.075} = \frac{1}{15},$$
- Q5: What is the conditional probability $p(\theta_2|x_3)$ that a patient has the disease θ_2 give that he has the symptom x_3?
  $$p(\theta_2|x_3) = \frac{p(x_3, \theta_2)}{p(x_3)} = \frac{0.005}{0.355} = \frac{p(x_3|\theta_2)p(\theta_2)}{p(x_3)} = \frac{\frac{1}{15} \times 0.075}{0.355} = \frac{1}{71}$$

More questions

- **Additional Question 1**: What is the probability of observing x_3 (symptom) given different values of Θ (disease)
- **Additional Question 2**: What are the conditional probabilities of a patient has all these disease given that he has the symptom x_3?

The Likelihood function

Answer to Additional Question 1: For this example, we use the likelihood function of observing x_3 given different values of Θ, defined as

$$\begin{aligned} p(x_3|\Theta) &= \{p(x_3|\theta_1), p(x_3|\theta_2), \ldots, p(x_3|\theta_{10})\} \\ &= \{0, \frac{1}{15}, \frac{1}{19}, \ldots, \frac{5}{9}\}, \end{aligned}$$

which can be written as

$$P(X = x|\Theta) = p(x|\Theta) \tag{6}$$

More Question: is this likelihood function the same as we learned in maximum likelihood method?

The Likelihood function

Question: is this likelihood function the same as we learned in the Maximum Likelihood Estimation (MLE)?

Answer: The same but with different notations and different interpretations. The likelihood function in MLE:

$$L(\theta|x) = P_X(x;\theta),$$

where $P_X(x;\theta)$ is the PMF of X parametrised by parameter θ

The Likelihood function: Frequentist vs Bayesian

Two interpretations from Frequentist and Bayesian of the Likelihood function, based on different interpretations of θ and Θ (although they are the same):

- Frequentist:
  - θ is the parameter of a predefined statistical model
  - The likelihood function is a statistical model that summarises a single random sample from a population [1], whose output value depends on a choice of parameters θ
- Bayesian:
  - Θ is a random variable
  - A function whose value is proportional to the probability of the conditional probability distribution X given Θ.

[1] Here "a single sample" means a specific subset of the population. For example, if we define the population as all patients admitted to QE hospital, a (random) sample could be defined as the patients aged 20-44 with trauma admitted in the past 1 months.

Bayes' Theorem for Discrete Distributions

Bayes' theorem for Probability Distributions: Given two random variables, X and Θ

$$p(\Theta|x) = \frac{p(\Theta)p(x|\Theta)}{p(x)} \tag{7}$$

where $p(\Theta|x)$ is called posterior distribution, $p(x|\Theta)$ is the likelihood function, $p(\Theta)$ is called the prior, x is a value of X and $x \in R_X$, and $p(x)$ is a scalar, called marginal likelihood.

Note: Marginal likelihood represents the probability of generating the observed sample for all possible values of the parameters (e.g., Θ), that is, the probability of the model (the likelihood function $p(x|\Theta)$) itself, or the evidence in favour of the model, so it's also known as model evidence or simply evidence.

In plain English:

$$\text{posterior distribution} = \frac{\text{prior distribution} \times \text{likelihood function}}{\text{marginal likelihood (model evidence)}}$$

Bayes' Theorem for Discrete Distributions

Additional Question 2 (pp.10): What are the conditional probabilities of a patient has all these diseases given that he has the symptom x_3?
Answer: similar to the calculation of $p(\theta_2|x_3)$ (Q5, pp.10), we have

$$
\begin{aligned}
p(\Theta|x_3) &= \frac{p(x_3|\Theta)p(\Theta)}{p(x_3)} \\
&= \{0, 1/71, 0.01, 0.14, 0.22, 0.15, 0.17, 0.01, 0.11, 0.07\}
\end{aligned}
$$

Question: What is the most possible disease given the patient has the symptom x_3?
Answer: We use **Maximum a posteriori probability (MAP) estimate** to the value of Θ that correspond to the maximum value of $p(\Theta|x_3)$, i.e.,

$$\hat{\theta}_{\mathsf{MAP}} = \underset{\theta}{\operatorname{argmax}}\, p(\Theta|x_3) = \theta_5$$

Question: what is the link between Maximum Likelihood estimation and MAP?

Summary and Further reading

- Probability and Statistics: Chapter 7. Bayesian Inference
- Probability for Computer Scientists, Standord Univ, CS109